Truck Turnaround Time (TAT) Analysis and Reduction in Warehouse Operations

Project report: Problem Statement, Business Importance, Key Assumptions in Data and Work Flow

1. Problem Statement

In FMCG distribution and warehousing, trucks spend a large part of their visit waiting rather than being loaded or unloaded. Truck Turnaround Time (TAT) is the total time from the moment a truck checks in at the security gate to the moment it exits the facility:

Total TAT = Exit Time - Arrival Time

TAT is made up of five sequential stages : gate-to-dock, dock waiting, loading/unloading, documentation and gate exit.

This project analyses 2,500 truck visits across 4 warehouses (Patna, Kolkata, Jamshedpur and Ranchi) over April to September 2026. The operational target is a TAT of 3.0 hours (180 minutes). The data shows a clear gap against that target:

- Average TAT is 4.32 hours (259 minutes) and the median is 4.02 hours.
- Only 21.08% of trucks (527 of 2,500) finish within 3.0 hours. 78.92% (1,973 trucks) breach the target.
- The average truck needs to save about 79 minutes to reach the target.
- Dock waiting averages 84 minutes (32% of TAT), and 49.4% of trucks wait more than 60 minutes.
- 59.3% of trucks were assigned to a dock still occupied by the previous truck. They waited 125.1 minutes on average, against 24.0 minutes when the dock was free.
- Dock Congestion is the most frequent delay reason (1,095 trips, 71.5% of flagged delays) and the most costly one (about 2,573 extra truck-hours).

Problem in one line: Trucks spend on average 4.32 hours per visit against a 3.0-hour target, mainly because of dock queues, and the project must identify where the time is lost and how much each stage must be cut to close the gap.

Objectives

1. Measure current TAT performance against the 3.0-hour benchmark.
2. Break TAT into stages and find the stages with the largest delay.
3. Compare performance across warehouses, months, arrival hours and transporters.
4. Identify the root causes of delay and quantify their impact in truck-hours.

5. Estimate how much improvement better dock allocation and stage-wise time cuts can deliver.

2. Business Importance

Reducing truck TAT matters because the yard and the docks are the bottleneck of every outbound and inbound flow.

- **More throughput without new infrastructure.** Docks blocked by waiting trucks cannot serve other vehicles. Shorter dwell time lets the same bays handle more freight without physical expansion. Patna, with 3 docks and about 4.7 trucks per dock per day, is the first candidate for relief.
- **Lower detention and yard cost.** Transporters charge detention when vehicles are held beyond contracted free hours. About 3,053 extra truck-hours were lost against the no-delay baseline, and 88.7% of them come from dock-wait related reasons.
- **Better service level (OTIF).** Delayed loading pushes dispatch late and risks missed On-Time In-Full delivery to distributors and retail points, which can lead to stockouts.
- **Stronger carrier relationships.** Long, unpredictable waits frustrate drivers and transporters and make it harder to secure reliable freight capacity.
- **Targeted action instead of general effort.** The analysis shows that gate and paperwork improvements alone cannot close the gap. Gate-to-dock (22.6 min), documentation (32.6 min) and gate exit (8.6 min) together are small compared with dock wait (84.0 min) and loading (111.6 min). Management effort is best spent on dock allocation, slot booking and dock capacity.
- **Quantified improvement potential.** Sending each truck to the first free dock in its warehouse would cut average dock wait from 84.0 to 45.8 minutes, TAT from 4.32 to 3.69 hours and compliance from 21.1% to 36.1%. This is a large gain, but it does not reach the target alone, so additional measures are also needed.

3. Key Assumptions in the Data

3.1 Nature and scope of the data

- **Synthetic data.** The dataset is sample operational data created for an academic portfolio, generated with a yard-queue simulation so that dock waiting, loading, documentation and delay reasons stay consistent with each other. The relationships found (dock busy vs free, workload per dock, hourly queue build-up) reflect the simulation and must be validated on real warehouse data.
- **Sampled period.** The 2,500 visits come from 42 sampled operating days (7 days per month, April to September 2026), about 60 trucks per day across the network. Monthly figures therefore show volume differences between sampled days, not a long-term trend of improvement or deterioration.

- **Calibration.** The simulation is calibrated to an average dock wait of about 84 minutes.
- **Four linked tables.** The truck visit data is joined with warehouse, transporter and dock master tables using ID columns, and the master tables are assumed to be correct and complete.

3.2 Business definitions and targets

- **3.0-hour benchmark.** The target is the dataset's own On Target threshold and is an assumed project target, not a company-confirmed SLA.
- **TAT definition.** TAT is calculated as Exit Time minus Arrival Time, and this recalculated value is treated as the true TAT. The original total_tat_hours column was checked against it (maximum difference below 0.05 hours) and then dropped.
- **Stage definitions.** Gate-to-dock, dock wait, loading/unloading, documentation and gate exit are treated as sequential, non-overlapping stages.
- **Peak hours.** Peak arrival hours are defined as 10:00-11:59 and 14:00-15:59 based on the arrival-hour chart.
- **Outliers.** A trip is flagged as a high-TAT outlier when TAT exceeds Q3 + 1.5 x IQR (8.07 hours). Outliers are flagged, not deleted, because they are real queueing events (all 82 flagged trips were logged as Dock Congestion).

3.3 Data quality assumptions

- **Timestamp format.** All milestone timestamps follow the dd-mm-yyyy HH:MM format and are converted to datetime. Seconds are not recorded, so durations are accurate to the minute.
- **Completeness.** The raw data has 2,500 records with no missing values and no duplicate rows.
- **Chronological order.** Milestones are assumed to follow the order: arrival, dock assigned, loading start, loading end, documentation complete, exit. Records are validated for this order, and for negative or impossible durations.
- **One delay reason per visit.** Each visit has a single primary delay reason. The label No Major Delay is used as the baseline, with an average TAT of 3.10 hours. Extra truck-hours for each reason are measured against this baseline.
- **No carrier-level cause data.** The data does not record why a transporter is slow, so explanations such as gate-registration delays or yard unfamiliarity are possible but not proven.

3.4 Analytical and modelling assumptions

- **Dock occupancy.** A dock is treated as occupied from dock wait through loading, and a truck is treated as arriving at a busy dock if the previous truck at that dock has not finished loading.
- **Dock utilisation.** Utilisation is calculated on a 24-hour dock day, but trucks arrive only between 08:00 and 19:59. Reported utilisation (for example 74% for Patna and 41% for Ranchi) therefore understates the real utilisation during working hours.

- **Dock allocation simulation.** The simulation assumes first-come, first-served order, each truck going to the dock that becomes free first within its own warehouse, and loading times, other stage times and truck arrival times staying unchanged. Only dock wait is recalculated.
- **Stage-wise improvement limits.** The best achievable level for each stage is assumed to be the network's own 25th percentile. Under this assumption the maximum possible saving is 112 minutes per truck, against the 79 minutes needed.
- **Optimisation costs.** The cost of cutting each stage in the optimisation scenarios is expressed in assumed relative cost units, not actual currency, so the plan costs are for comparison between scenarios only.
- **Statistical testing.** The spot vs contract carrier comparison uses a Welch t-test ($p = 0.12$), and the 6-minute gap is not statistically significant.

4. Project Work Flow

The project was carried out in ten steps, from problem definition to final recommendations, using Python (Pandas, NumPy, Matplotlib, Seaborn).

1. **Define the problem and target.** Fix the scope (4 warehouses, 2,500 truck visits, April to September 2026), define TAT and its five stages, and set the 3.0-hour benchmark.
2. **Load the data.** Import the truck visit data and the three master tables (warehouse, transporter, dock) into Pandas DataFrames and check the shape of each.
3. **Understand the raw data.** Check column names, data types, duplicate rows, missing values and summary statistics. Finding: all timestamps are stored as text and must be converted.
4. **Clean and validate.** Convert timestamps to datetime, check the chronological order of milestones, check for negative durations, recalculate TAT from arrival and exit times and compare it with the raw column, and flag outliers with the IQR method.
5. **Create new columns.** Add benchmark compliance (within or above 3.0 hours), peak-hour flag and arrival weekday, drop duplicate or unused columns, and save the cleaned file.
6. **Merge with master tables.** Join the cleaned visit data with warehouse, transporter and dock details so that results can be compared by facility and carrier.
7. **Exploratory data analysis.** Answer eight business questions: overall TAT and SLA adherence, TAT distribution, monthly trend, stage-wise breakdown, warehouse comparison, peak arrival hours, delay root causes and transporter performance.
8. **Root-cause and workload analysis.** Compare trucks arriving at a busy dock with those arriving at a free dock, and compare workload per dock across warehouses. Finding: dock queues are the main driver of delay.
9. **Optimisation in two layers.** Layer 1 simulates sending each truck to the first free dock in its warehouse (TAT falls from 4.32 to 3.69 hours). Layer 2 works out how many minutes each stage must be cut (79 minutes needed against 112 minutes

maximum possible) and compares the cost of different cut plans under three scenarios.

10. **Insights and recommendations.** Summarise findings, recommend dock allocation, slot booking and extra dock capacity (starting with Patna), and state the data limitations.

Flow in one line: Problem and target, then data loading, raw data check, cleaning and validation, feature creation, merging, EDA, root-cause analysis, optimisation, and recommendations.